# Bayesian Rationality as an Identifying Restriction

Preliminary and incomplete

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**Abstract**

When feedback is coarse, a player may learn about the distribution of her consequences without disentangling how fundamentals and an opponent's behavior jointly generate those consequences. Berk–Nash equilibrium disciplines beliefs by statistical fit and actions by optimality, but a best-fitting hypothesis may attribute to an opponent a strategy that is not optimal in the problem represented by that hypothesis. This paper introduce cross-player Bayesian rationality: the strategy attributed to an opponent must be optimal under one belief of that opponent concentrated on the parameters that best fit her data in the corresponding represented problem. Iterating this condition imposes progressively stronger consistency requirements on Berk–Nash explanations. The framework separates behavioral selection from identification of the state–strategy hypotheses supporting behavior.

**Keywords:** Berk–Nash equilibrium, misspecified games, identification, conjectures, rationality, self-confirming equilibrium, rationalizable conjectural equilibrium.
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## 1 Introduction

A player's data may identify the distribution of her consequences without identifying whether those consequences were generated by nature or by an opponent's behavior. These explanations can agree about observations following the player's chosen actions while implying different counterfactual consequences and different accounts of how the opponent would act. Berk–Nash equilibrium disciplines each player's belief through her observed data and requires her strategy to be optimal under that belief (Esponda and Pouzo, 2016). We ask what further restrictions follow when a fitted hypothesis must attribute to the opponent a strategy that she can support using a best-fitting belief in the represented decision problem. We call this requirement cross-player Bayesian rationality.

Three examples develop the question. In an investment game, coarse feedback permits an outside option to be supported by conjectures assigning positive probability to an action that is strictly dominated for the opponent. Evaluating the attributed behavior removes those conjectures and then the outside option, which is not itself strictly dominated in the original game. A matching-distance game shows that dominance is not the mechanism: no location is dominated, but distance feedback permits mismatched Berk–Nash profiles. Optimality alone does not eliminate them; they are eliminated when the belief supporting each attributed strategy must also fit the observations generated in the represented problem.

The coordination example adds an unknown payoff-relevant state. A player's observations identify a composite of the state distribution and the opponent's behavior, so several state–strategy hypotheses fit the same data. These hypotheses imply different observations and incentives for the represented opponent. The first cross-player application eliminates one Berk–Nash profile even though some fitted hypotheses satisfy the opponent's fit-and-optimality test, because no belief over those hypotheses supports the original player's action. At another profile, behavior is unchanged but the supporting belief becomes unique. Thus the same data can eliminate one profile and pin down the belief supporting another.

A parameter $(\pi^i, \widehat{\sigma}^j)$ is one hypothesis held by player $i$: Nature draws the state according to $\pi^i$, while player $j$ behaves according to $\widehat{\sigma}^j$. It is not player $j$'s belief. In the represented problem generated by this hypothesis and player $i$'s strategy, player $j$ uses her own subjective model and her own belief. That belief must be concentrated on the parameters that best fit her represented data and must support the strategy $\widehat{\sigma}^j$. Player $i$ may then place weight only on her own best-fitting hypotheses that meet this requirement.

The belief supporting player $j$'s attributed strategy may in turn contain hypotheses about player $i$'s behavior. The paper therefore applies the same test through successive represented problems. Berk–Nash equilibrium is denoted by $E^0$, the first cross-player application gives $E^1$, and $E^\infty$ imposes the test at every finite order. These sets are nested. The recursion imposes increasingly deep mutual consistency of statistical fit and optimal choice; it does not construct epistemic types, literal common knowledge, or a common prior.

The specification analysis distinguishes truth admission, reproduction of observed conditional distributions, and identification. A false state–strategy hypothesis can reproduce the data, while admitting the objective state distribution need not identify either component of a hypothesis. Two hypotheses that are observationally equivalent for player $i$ may also generate different represented problems for player $j$. The first cross-player application may therefore remove some observationally equivalent hypotheses, narrow the beliefs supporting an unchanged profile, or eliminate the profile. Identification is conditional on a fixed profile and the maintained subjective models; it does not imply unique behavior or posterior convergence.

The binary analysis gives an exact one-dimensional payoff-gap formulation without restricting feedback. With two actions, the next-order condition holds exactly when the convex hull of payoff differences generated by best-fitting hypotheses whose attributed strategies satisfy the preceding-order restriction intersects the sign region required by optimality. This exposes two sources of failure: either no best-fitting hypothesis satisfies the earlier restriction, or the hypotheses that do cannot jointly support the proposed strategy.

The section then gives sufficient conditions for existence at every order. Bilateral truth admission guarantees that every objective Nash equilibrium belongs to $E^\infty$, but it need not make $E^\infty$ equal to $E^0$, identify the supporting beliefs, or rule out additional profiles. More generally, a Nash equilibrium under possibly different state distributions belongs to $E^\infty$ if each player's selected state–strategy hypothesis best fits both her physical data and the data in the represented problem generated by the other player's state distribution. This condition permits misspecification and positive fitting loss but is not necessary, since a mixture of best-fitting hypotheses may support the same behavior. The analysis is static and does not assert that the supporting beliefs arise as limits of Bayesian learning.

### 1.1 Relation to the Literature

The paper takes the fitting-and-optimality discipline from Esponda and Pouzo (2016). Its state–strategy parameterization specializes that framework by making explicit one possible state distribution and one strategy attributed to the opponent. This decomposition permits the attributed strategy to be evaluated in a represented problem without treating the parameter as the opponent's belief. The complete-information examples also connect to self-confirming equilibrium (Fudenberg and Levine, 1993): coarse feedback permits conjectures that reproduce observed consequences while being wrong about unobserved behavior. That analogy requires exact on-path fit, whereas the present analysis also permits positive minimum loss.

The closest strategic comparisons are rationalizability and rationalizable conjectural equilibrium. Rationalizability iteratively removes behavior that is not optimal under any belief about opponents' remaining behavior (Bernheim, 1984; Pearce, 1984; Battigalli and Siniscalchi, 2003); here the belief supporting an attributed strategy must also be concentrated on parameters that best fit the represented data. Rubinstein and Wolinsky (1994) combine optimality, feedback consistency, and recursive conjectures in complete-information games, while Esponda (2013) studies rationalizable conjectural equilibrium with structural and strategic uncertainty. The matching-distance example exhibits their common fitting-and-choice logic, but the present paper does not assert a general equivalence. Esponda and Pouzo (2025) instead use Berk–Nash rationalizability to characterize long-run actions in complete-information games with misspecified models. The present recursion evaluates the strategy component inside fitted state–strategy hypotheses at a fixed physical profile and permits uncertainty about the state.

Section 2 develops the three examples, and Section 3 introduces the environment and Berk–Nash equilibrium. Section 4 defines the represented problem and the recursive cross-player restriction. Section 5 separates specification from identification and studies the first cross-player application. Section 6 gives the payoff-gap formulation for binary games and sufficient conditions for existence at every order.

The identification objective is whether, conditional on a fixed profile $\sigma \in E^{\infty}$, each actual player has a unique supporting belief over her own state–strategy parameters. Complete uniqueness concerns the entire probability measure; on-path uniqueness concerns only its pushforward under the likelihood-relevant prediction map. The current draft defines the recursive restrictions and illustrates how they can narrow or identify a supporting belief, but the general characterization of belief uniqueness under arbitrary feedback is not yet complete. The binary results that follow therefore provide conditions for $\sigma \in E^{\infty}$ and for $E^{\infty} \neq \varnothing$, not a general theorem identifying supporting beliefs.

## 2 Motivating Examples

Berk–Nash equilibrium disciplines each player's belief through her observed data and requires her strategy to be optimal under that belief (Esponda and Pouzo, 2016). When a player's subjective parameter contains a conjecture about an opponent's strategy, Berk–Nash equilibrium evaluates that parameter only through the consequences it predicts for the player holding it.[1] We study the further restrictions obtained when the behavior attributed within a fitted parameter is evaluated in the opponent's represented problem using the same fitting-and-optimality criterion.

The first example shows what follows from imposing the opponent's optimality condition alone. Coarse feedback permits a Berk–Nash equilibrium to rely on conjectures assigning positive probability to an action that is dominated for the opponent. Because a dominated action cannot be optimal under any belief, requiring the attributed strategy to be optimal for the opponent rules out those conjectures and the equilibrium they support. The second example is a complete-information game without dominated actions. A conjecture may fit a player's coarse feedback. Viewed from the opponent's perspective, however, the represented environment and the strategy attributed to her induce a distribution over the observations she would receive. The attributed strategy may not be optimal under any belief that fits those observations. We therefore impose the additional epistemic assumption that each player regards the opponent as Bayesian rational: the opponent uses her own observations to discipline her belief and chooses optimally under that belief. In the example, this requirement eliminates the non-Nash Berk–Nash equilibria. The example also motivates the connection between our concept and the rationalizable conjectural equilibrium of Rubinstein and Wolinsky (1994) in complete-information games.

[1] In the examples below, the minimum weighted Kullback–Leibler divergence is zero, so fit can be stated as exact reproduction of the relevant consequence distributions. Section 3 gives the general definition.

The third and main example adds an unknown payoff-relevant state to a coordination game. A player's feedback depends jointly on nature and the opponent's behavior: the same observation can be explained by a favorable state occurring more often or by the opponent coordinating more often. In this example, every attributed strategy is optimal under some unrestricted belief, so optimality alone cannot separate the roles of nature and behavior. Imposing the same epistemic assumption—that each player regards the opponent as Bayesian rational—eliminates one Berk–Nash equilibrium and, at another profile, uniquely identifies each actual player's supporting belief. Some explanations satisfy the first application but do not satisfy subsequent applications, which motivates applying the restriction recursively.

**Example 1** (Investment game)**.** Player 1 chooses whether to invest, $I$, or take an outside option, $O$; player 2 chooses whether to assist, $A$, or block, $B$. If player 1 invests, her consequence is success when player 2 assists and failure when player 2 blocks. If she takes the outside option, her consequence is no project regardless of player 2's action. Player 2's consequence is assisting after $A$ and blocking after $B$.

Player 1 receives payoff 2 from success, 0 from failure, and 1 from no project. Player 2 receives payoff 1 from assisting and 0 from blocking. The payoff matrix is therefore

|  | $A$ | $B$ |
|---|---|---|
| $I$ | $(2,1)$ | $(0,0)$ |
| $O$ | $(1,1)$ | $(1,0)$. |

Each player observes her own action and consequence but not the opponent's action.

There is no uncertainty from nature, so we suppress the degenerate state-distribution component. A player's parameter can therefore be identified with a conjecture about the opponent's strategy, and each player's subjective model includes the whole strategy space. The action sets, consequence and payoff functions, feedback structures, and subjective model classes are common knowledge. Each player can therefore evaluate whether a strategy attributed to the opponent could be optimal in the opponent's decision problem. This does not imply that either player knows the opponent's actual strategy or belief.

Player 2's action $A$ strictly dominates $B$. After $B$ is deleted, player 1 strictly prefers $I$. The game is therefore dominance solvable with its unique Nash equilibrium $(I, A)$. Because each player admits the opponent's actual strategy, point beliefs on actual play fit exactly and support both strategies, so $(I, A)$ is also a Berk–Nash equilibrium.

Berk–Nash equilibrium nevertheless also admits $(O, A)$. At $(O, A)$, player 1 observes no project. Since this consequence occurs after $O$ whether player 2 assists or blocks, every conjecture about player 2 fits. In particular, if player 1 is certain that player 2 chooses $B$, investment yields 0, whereas the outside option yields 1. The point belief on this conjecture therefore supports $O$. Because $A$ is strictly optimal for player 2 under every belief, player 2 chooses $A$. Consequently, the Berk–Nash equilibria are[2]

$$(I, A) \qquad \text{and} \qquad (O, A).$$

The profile $(O, A)$ relies on behavior attributed to player 2 that cannot be optimal under any belief. Indeed, any belief supporting $O$ must assign positive probability to the set of conjectures under which player 2 chooses $B$ with probability at least 1/2.

Suppose now that player 1 regards player 2 as rational, in the minimal sense that every strategy she attributes to player 2 must be optimal for player 2 under some belief. Because $A$ strictly dominates $B$, no strategy assigning positive probability to $B$ can be optimal. Rationality therefore rules out every conjecture needed to support $O$.

The only conjecture consistent with player 2's rationality is pure $A$. Under that conjecture, player 1 strictly prefers $I$, so no belief over rational conjectures supports $O$. Requiring opponent rationality therefore eliminates $(O, A)$ and leaves the dominance-solvable outcome $(I, A)$. □

**Example 2** (Matching distance)**.** Two players simultaneously choose locations $x^i \in \{0, 1, 2, 3\}$. Player $i$'s consequence is the distance $y^i = |x^i - x^j|$, and her payoff is $u^i(x^i, y^i) = -(y^i)^2$. Each player observes her own location and the resulting distance but does not observe the opponent's location. This is a finite version of the distance game in Rubinstein and Wolinsky (1994).

There is no uncertainty from nature, so we suppress the degenerate state-distribution component. A player's parameter is simply a conjecture about the opponent's strategy, and every such conjecture is considered. The primitives and subjective model classes are common knowledge. Under quadratic distance loss, a best response to a conjecture is a feasible location closest to its mean.

The Nash equilibria are the four matching pure profiles and the three profiles in which both players mix equally between 0 and 1, between 1 and 2, or between 2 and 3. The profile $(1, 2)$ is not Nash because either player can move to the other's location and increase her payoff from $-1$ to 0.

Consider nevertheless $(1, 2)$. Player 1 chooses location 1 and observes distance 1. Her point belief on the conjecture assigning probability 1/2 to each of locations 0 and 2 fits this observation and makes location 1 uniquely optimal. Player 2 also observes distance 1. Her point belief on the conjecture assigning probability 1/2 to each of locations 1 and 3 fits this observation and makes location 2 uniquely optimal. Thus $(1, 2)$ is a Berk–Nash equilibrium. Reversing the players' locations gives the additional Berk–Nash equilibrium $(2, 1)$. The Berk–Nash equilibrium set consists of the seven Nash equilibria together with these two profiles.[3]

The two additional profiles have the self-confirming form. Each player holds an incorrect conjecture about the opponent's location, but the conjecture correctly predicts the distance following her chosen action. The subjective model classes are correctly specified because each contains the

[2] There are no additional Berk–Nash equilibria. Player 2 must choose $A$. If player 1 assigns positive probability to $I$, success follows $I$ with probability one. Exact fit then requires every conjecture receiving positive weight in her belief to assign probability one to $A$. Under every such conjecture, $I$ is strictly better than $O$, so player 1 cannot mix.

[3] There are no additional Berk–Nash equilibria. At every other mismatched pure profile, at least one player's distance identifies the opponent's location, which she strictly prefers to match. If a player mixes, a sampled boundary location identifies the opponent's strategy, while the distance distributions following both interior locations jointly identify it. A player cannot mix against a pure opponent because matching is uniquely optimal. Hence every Berk–Nash equilibrium involving mixing is Nash.

opponent's actual strategy. The difference between Nash and Berk–Nash equilibrium therefore comes from incomplete feedback rather than misspecification (Fudenberg and Levine, 1993; Esponda and Pouzo, 2016).

First impose the minimal rationality condition introduced in the investment example: player $i$'s supporting belief may assign positive probability only to conjectures under which the strategy attributed to the opponent is optimal under some belief the opponent could hold. At $(1, 2)$, every conjecture fitting player 1's observation has support contained in $\{0, 2\}$. A conjecture assigning positive probability to both locations cannot be optimal for player 2. For locations 0 and 2 both to be optimal, the mean of player 1's location would have to be 1, but player 2 would then strictly prefer location 1 to either endpoint.

Rationality therefore permits only the conjecture assigning probability one to location 0 and the conjecture assigning probability one to location 2. Each attributed strategy is optimal for player 2 when she believes that player 1 chooses the same location. Although the point belief on the mixed conjecture used above attributes a strategy that is not optimal under any belief, player 1 can instead assign probability $1/2$ to each of these two conjectures. This belief gives player 1 the same expected payoff from every location as the point belief on the mixed conjecture and therefore still makes location 1 uniquely optimal. Each conjecture receiving positive probability attributes a strategy that is optimal for player 2 under some belief. The analogous belief for player 2 supports location 2. Thus rationality alone does not eliminate $(1, 2)$.

Suppose now that each player regards the opponent as Bayesian rational. Consider a conjecture of player 1 that attributes a particular location to player 2. Player 1's actual location and the location attributed to player 2 determine the distance that player 2 would observe in the represented problem. Player 1 asks whether there exists a belief of player 2 over conjectures about player 1 that both fits this represented observation and makes the attributed location a best response. This belief belongs to player 2 in the represented problem and is distinct from player 1's supporting belief over conjectures about player 2. If such a belief exists, the conjecture is consistent with player 2's Bayesian rationality; otherwise, it is ruled out.

Consider first player 1's conjecture that player 2 chooses location 0 with certainty. Given player 1's actual location 1, player 2 would observe distance 1 at the boundary location 0. This observation identifies player 1's location as 1. Every belief of player 2 that fits this observation therefore makes location 1 strictly better than the attributed location 0. The conjecture fails the represented fitting-and-optimality requirement.

Now consider player 1's conjecture that player 2 chooses location 2 with certainty. Player 2 would again observe distance 1, which is consistent with player 1 choosing either location 1 or location 3. Player 2's point belief on the conjecture assigning equal probability to these two locations fits that observation and makes location 2 uniquely optimal. This conjecture therefore satisfies the requirement.

Consequently, player 1's supporting belief must be concentrated on the conjecture that player 2 chooses location 2 with certainty. Such a belief cannot support player 1's choice of location 1, since moving to 2 raises her payoff from $-1$ to 0.

For player 2, the same test rules out the conjecture that player 1 chooses location 3 with certainty and leaves only the conjecture that player 1 chooses location 1 with certainty. A belief concentrated on that conjecture cannot support player 2's choice of location 2. Thus the additional cross-player Bayesian-rationality requirement fails in both directions, eliminating $(1, 2)$ and, with the players' locations reversed, $(2, 1)$. At each of the seven Nash equilibria, point beliefs on actual play satisfy fitting and optimality in both represented problems, so all seven satisfy the requirement.

The example separates the two requirements. Rationality alone rules out fitting conjectures under which the opponent assigns positive probability to both nonadjacent locations. It does not rule out a belief over the remaining pure conjectures: each attributed pure strategy is optimal under some belief, while their average continues to support the non-Nash profile. The additional force comes from requiring the belief supporting each attributed strategy to fit the feedback generated in the represented player's decision problem. Unlike in the investment example, no conjectured location is dominated. The restriction arises from the interaction of fitting and optimality in the represented problem.

Rubinstein and Wolinsky (1994) define rationalizable conjectural equilibrium by requiring each player's action to be optimal under a conjecture consistent with her observed signal, while every opponent action–signal pair used by that conjecture must itself be rationalizable in the same way. In complete-information games, our recursive fitting-and-optimality requirement imposes the same reasoning on the behavior represented within each player's parameter. The results below do not require an equivalence between the two constructions. □

**Example 3** (Coordination with state uncertainty)**.** Two symmetric players, $i$ and $j$, choose $L$ or $R$. A state $\omega \in \{\alpha, \beta\}$ is drawn according to the objective distribution $p$, where $p(\alpha) = 1/2$. A player's consequence is high if and only if the two actions coincide and $\omega = \alpha$, and low otherwise. There is no private information before actions are chosen. After play, each player observes her own action, consequence, and payoff, but neither the state nor the other player's action. Payoffs are

| | high | low |
|---|---|---|
| $L$ | 3 | 0 |
| $R$ | $\frac{1}{2}$ | 2 |

Each player's subjective model considers the state distributions $\pi^i(\alpha) \in \{1/2, 2/3, 1\}$ and every strategy of the opponent. A parameter $(\pi^i, \widehat{\sigma}^j)$ is one hypothesis held by player $i$: nature draws the state according to $\pi^i$, while player $j$ behaves according to $\widehat{\sigma}^j$. Under this parameter, a high consequence occurs after $L$ with probability $\pi^i(\alpha)\widehat{\sigma}^j(L)$ and after $R$ with probability $\pi^i(\alpha)\widehat{\sigma}^j(R)$. Expected payoffs are

$$U^i\big(L; (\pi^i, \widehat{\sigma}^j)\big) = 3\pi^i(\alpha)\widehat{\sigma}^j(L), \qquad U^i\big(R; (\pi^i, \widehat{\sigma}^j)\big) = 2 - \frac{3}{2}\pi^i(\alpha)\widehat{\sigma}^j(R).$$

At the objective state distribution, the payoff difference is $U^i(L; (p, \widehat{\sigma}^j)) - U^i(R; (p, \widehat{\sigma}^j)) = \frac{3}{4}\widehat{\sigma}^j(L) -$

$\frac{5}{4} < 0$ for every $\widehat{\sigma}^j$. Thus $R$ strictly dominates $L$ in the objective game, and the unique Nash equilibrium is $(R, R)$.

Berk–Nash equilibrium also includes $(L, L)$. A player choosing $L$ observes a high consequence with probability $1/2$, so exact fit requires $\pi^i(\alpha)\widehat{\sigma}^j(L) = 1/2$. The data reveal how often state $\alpha$ and the opponent's action $L$ occur together, but not their separate frequencies. The fitting pairs $(\pi^i(\alpha), \widehat{\sigma}^j(L))$ are $(1/2, 1)$, $(2/3, 3/4)$, and $(1, 1/2)$. Their payoff differences $U^i(L) - U^i(R)$ are, respectively, $-1/2$, $-1/4$, and $1/4$. The point belief on $(1, 1/2)$ supports $L$, and every fitting belief supporting $L$ must assign positive probability to this parameter.

At $(R, R)$, exact fit requires $\pi^i(\alpha)\widehat{\sigma}^j(R) = 1/2$. The fitting pairs are $(1/2, 0)$, $(2/3, 1/4)$, and $(1, 1/2)$. Every belief of player $i$ concentrated on the first two parameters fits her data and supports her action $R$. By symmetry, both players have supporting beliefs of this form, so $(R, R)$ is a Berk–Nash equilibrium.

No other profiles are Berk–Nash equilibria.[4] Therefore,

$$E^0 = \{(L, L), (R, R)\}.$$

The hypothesis $(1/2, 1)$ at $(L, L)$ uses the objective state distribution and the opponent's actual action, yet it makes $R$ strictly better than $L$. Supporting $L$ requires a different account of the same observation: state $\alpha$ occurs more often and the opponent chooses $L$ less often. These hypotheses agree about the data observed after $L$ but disagree about what would happen after $R$.

Optimality of the strategy attributed to player $j$ does not separate these hypotheses. Player $j$'s point belief on $(\pi^j(\alpha), \widehat{\sigma}^i(L)) = (2/3, 1)$ gives $U^j(L) = U^j(R) = 2$. Since the two actions are tied, every strategy attributed to player $j$ is optimal under this belief.

Suppose now that player $i$ regards player $j$ as Bayesian rational. For every hypothesis receiving probability in player $i$'s supporting belief, the strategy attributed to player $j$ must be supported by a belief fitting the observations that player $j$ would receive in the represented environment. The state distribution in player $i$'s hypothesis and player $i$'s actual strategy determine those observations. The attributed strategy determines which actions player $j$ samples. The belief used to fit these observations belongs to player $j$ and is a belief over her parameters $(\pi^j, \widehat{\sigma}^i)$.

At $(L, L)$, any belief supporting player $i$'s choice must assign positive probability to $(1, 1/2)$. This parameter attributes probability $1/2$ to each action of player $j$. Because player $j$ samples both actions, her represented observations identify $(\pi^j(\alpha), \widehat{\sigma}^i(L)) = (1, 1)$: state $\alpha$ occurs with probability one and player $i$ chooses $L$. Under this parameter, player $j$ receives 3 from $L$ and 2 from $R$. No fitting belief supports the strategy attributed to her, which assigns positive probability to both actions. Consequently, no supporting belief of player $i$ satisfying the cross-player condition can assign positive probability to $(1, 1/2)$. The other two fitting parameters both make $R$ strictly better

[4]At an asymmetric pure profile, the player choosing $L$ observes no high consequence. Since every state distribution in her subjective model assigns positive probability to $\alpha$, exact fit requires her to attribute probability one to $R$ to the opponent, under which she strictly prefers $R$. If a player mixes, the conditional distributions of consequences following both actions identify the objective state distribution and the opponent's actual strategy. At that state distribution, $R$ is strictly optimal, so mixing is impossible.

than $L$, so no belief concentrated on them supports player $i$'s actual choice. Symmetry gives the same conclusion for player $j$, and $(L, L) \notin E^1$.

At $(R, R)$, the parameter $(1/2, 0)$ attributes $R$ with probability one to player $j$. Player $j$'s point belief on $(\pi^j(\alpha), \widehat{\sigma}^i(L)) = (1/2, 0)$ fits her represented observation and makes $R$ strictly optimal. The other two fitting parameters attribute positive probability to both actions. Player $j$ then samples both actions, and her observations identify that player $i$ chooses $R$ as well as the state probability. The resulting fitting parameter is $(2/3, 0)$ when player $i$'s hypothesis is $(2/3, 1/4)$, and $(1, 0)$ when it is $(1, 1/2)$. Both make $R$ strictly better than $L$, so neither supports the mixed strategy attributed to player $j$.

Any belief of player $i$ satisfying the cross-player condition must therefore assign probability one to $(1/2, 0)$. Its point belief supports $R$, so $\delta_{(1/2,0)}$ is player $i$'s unique belief supporting actual play after the first cross-player application. The same holds for player $j$. This uniqueness concerns the beliefs of the actual players; beliefs used within represented problems need not be unique.

Although $(L, L)$ is already excluded from $E^1$, its fitted parameters distinguish one application from full recursion. The parameter $(1/2, 1)$ can satisfy the first represented problem only through a belief assigning positive probability to $(1, 1/2)$. Since $(1, 1/2)$ cannot satisfy the condition in the next represented problem, $(1/2, 1)$ is removed at the following level. The parameter $(2/3, 3/4)$ satisfies the condition through two levels but not through a third; Appendix A.3 gives the calculation.

At $(R, R)$, the parameter $(1/2, 0)$ generates the same represented problem at every order and attributes probability one to $R$. Its point belief fits the represented observations and supports $R$ at every finite order. Hence

$$E^\infty = \{(R, R)\},$$

and $\delta_{(1/2,0)}$ is each actual player's unique supporting belief after the first cross-player application and at every higher order. $\square$

The coordination game explains why a parameter is treated as a hypothesis about a data-generating process. Two parameters can predict the same observations for player $i$ while assigning different roles to nature and player $j$'s behavior. Together with player $i$'s actual strategy, they can then imply different observations and incentives for player $j$. Player $i$ holds the parameter, whereas player $j$ holds the belief supporting the strategy attributed to her.

These differences can restrict both behavior and its explanation. At $(L, L)$, every belief supporting $L$ assigns positive probability to a parameter under which no belief fitting the opponent's represented observations supports the strategy attributed to her. At $(R, R)$, each actual player's unique supporting belief is $\delta_{(1/2,0)}$. One application does not assess the parameters used within the represented player's belief. The formal construction therefore applies the same fitting-and-optimality condition through successive represented problems.

# 3 Model and Berk–Nash Equilibrium

The three examples have the same underlying form: each player observes the consequences following her actions and interprets them through a subjective model. The maintained behavioral premise is Bayesian rationality: her belief is concentrated on the parameters that best fit these observations, and her strategy is optimal under that belief. Berk–Nash equilibrium applies these requirements separately to the two actual players. The primitives below distinguish the objective process that generates each player's consequences from the subjective models she uses to explain them.

## 3.1 Objective environment

There are two players, $I = \{1, 2\}$. Player $i \in I$ has a finite action set $X^i$ and a finite consequence set $Y^i$. Let $X = X^1 \times X^2$ and $Y = Y^1 \times Y^2$. A state $\omega \in \Omega$ is drawn according to an objective distribution $p \in \Delta(\Omega)$ with full support, where $\Omega$ is finite. Players receive no private information before choosing actions simultaneously. For distinct players $i$ and $j$, player $i$'s feedback function is

$$f^i : X^i \times X^j \times \Omega \to Y^i.$$

Following actions $(x^i, x^j)$ and state $\omega$, player $i$ observes the consequence

$$y^i = f^i(x^i, x^j, \omega) \in Y^i$$

and receives payoff $u^i(x^i, y^i) \in \mathbb{R}$. Apart from her own action, $y^i$ is player $i$'s only feedback.[5] Let $f = (f^i)_{i \in I}$ and $u = (u^i)_{i \in I}$. The underlying structure of the game is

$$\mathcal{S} = \langle I, \Omega, X, Y, f, u \rangle.$$

Together with the objective state distribution $p$, it defines the objective game

$$\mathcal{O} = \langle \mathcal{S}, p \rangle.$$

A strategy of player $i$ is $\sigma^i \in \Sigma^i \equiv \Delta(X^i)$. Let $\Sigma = \Sigma^1 \times \Sigma^2$ and denote a strategy profile by $\sigma = (\sigma^1, \sigma^2) \in \Sigma$. Fix an objective game. For each strategy profile $\sigma$, an objective distribution over player $i$'s consequences is given by

$$Q^i_\sigma(y^i \mid x^i) = \sum_{\omega \in \Omega} \sum_{x^j \in X^j} p(\omega) \sigma^j(x^j) \mathbf{1}\{f^i(x^i, x^j, \omega) = y^i\}.$$

This definition applies to every $(x^i, y^i) \in X^i \times Y^i$.

[5] Any additional randomness in consequences can be incorporated into the state $\omega$.

### 3.2 Decomposed subjective models

Following the structural interpretation in Esponda and Pouzo (2016), player $i$'s subjective parameter is

$$\theta^i = (\pi^i, \widehat{\sigma}^j) \in \Theta^i \equiv \Pi^i \times \Sigma^j,$$

where $\Pi^i \subseteq \Delta(\Omega)$ is the set of state distributions considered possible by player $i$ and $\widehat{\sigma}^j \in \Sigma^j$ is her conjecture about player $j$'s strategy.[6] By definition, player $i$ may rule out some state distributions, but she considers every strategy of player $j$ possible. We reserve $p$ for the objective state distribution. Within a parameter $\theta^i = (\pi^i, \widehat{\sigma}^j)$, the subjective consequence distribution is given by

$$Q^i_{\theta^i}(y^i \mid x^i) = \sum_{\omega\in\Omega} \sum_{x^j \in X^j} \pi^i(\omega)\widehat{\sigma}^j(x^j)\mathbf{1}\{f^i(x^i, x^j, \omega) = y^i\}.$$

For later use, extend this notation to any $\rho \in \Delta(\Omega)$ and any $\tau^j \in \Sigma^j$ by defining $Q^i_{\rho,\tau^j}(y^i \mid x^i) = \sum_{\omega\in\Omega} \sum_{x^j\in X^j} \rho(\omega)\tau^j(x^j)\mathbf{1}\{f^i(x^i, x^j, \omega) = y^i\}$.

Let $\Theta = \Theta^1 \times \Theta^2$ and $Q_\theta = (Q^i_{\theta^i})_{i\in I}$. The subjective model is

$$\mathcal{Q} = \langle \Theta, (Q_\theta)_{\theta\in\Theta}\rangle.$$

A game is composed of the objective game $\mathcal{O}$ and the subjective model $\mathcal{Q}$:

$$\mathcal{G} = \langle \mathcal{O}, \mathcal{Q}\rangle.$$

We assume that the underlying structure $\mathcal{S}$ and the subjective model $\mathcal{Q}$ are common knowledge. The objective distribution $p$ need not be known to the players, and each player's actual belief and strategy need not be known to the other player. In particular, player $i$ need not know player $j$'s actual belief or strategy.

**Interpretation of a subjective parameter.** A parameter $\theta^i = (\pi^i, \widehat{\sigma}^j)$ is one hypothesis in player $i$'s subjective model about how her consequences are generated. Under this hypothesis, Nature draws the state according to $\pi^i$ and player $j$ behaves according to the strategy $\widehat{\sigma}^j$. Together with the consequence function $f^i$, these components induce the conditional distribution $Q^i_{\theta^i}(\cdot \mid x^i)$ following each action $x^i$.

This decomposition gives player $i$'s subjective model a structural interpretation: each conditional distribution of consequences is induced by a possible distribution of the state and possible behavior by the opponent through the underlying consequence function. The parameter belongs to player $i$'s subjective model, and $\widehat{\sigma}^j$ is behavior that player $i$ considers possible for player $j$; it does not describe player $j$'s own subjective model or belief. Distinct parameters may induce the same conditional distribution of player $i$'s consequences, so the same distribution can have different

[6] Within a parameter, the state and player $j$'s current action are independent. This is consistent with the absence of information about the current state before actions are chosen. A belief over parameters need not factor, however. Player $i$ may regard different state distributions as being associated with different steady-state strategies of player $j$.

structural explanations in terms of the state distribution, the behavior attributed to player $j$, or both.

### 3.3 Berk–Nash equilibrium (Esponda and Pouzo (2016))

Berk–Nash equilibrium disciplines each player's belief through the consequences she observes following her actions and requires her strategy to be optimal under that belief (Esponda and Pouzo, 2016). We define statistical fit and optimality in turn.

**Statistical fit.** Fix a strategy profile $\sigma$. Conditional on player $i$ choosing action $x^i$, the actual distribution of her consequence is $Q^i_\sigma(\cdot \mid x^i)$, while parameter $\theta^i$ predicts $Q^i_{\theta^i}(\cdot \mid x^i)$. The distance between these distributions is measured by the weighted Kullback–Leibler divergence

$$K^i(\sigma, \theta^i) = \sum_{x^i \in X^i} \sigma^i(x^i) \sum_{y^i \in Y^i} Q^i_\sigma(y^i \mid x^i) \ln \frac{Q^i_\sigma(y^i \mid x^i)}{Q^i_{\theta^i}(y^i \mid x^i)}.$$

We use the conventions $0 \ln(0/q) = 0$ and $q \ln(q/0) = +\infty$ for $q > 0$. Let

$$\Theta^i(\sigma) \equiv \arg\min_{\theta^i \in \Theta^i} K^i(\sigma, \theta^i)$$

denote the set of parameters that best fit the data generated at $\sigma$. Only actions in $\operatorname{supp}(\sigma^i)$ receive positive weight in the fitting criterion. Consequently, statistical fit disciplines the conditional distributions following actions that player $i$ plays but does not restrict predictions following her unplayed actions.

**Optimal decision.** Each parameter predicts a conditional distribution of consequences following every action, including actions that player $i$ does not play at $\sigma$. The expected payoff from action $x^i$ under parameter $\theta^i \in \Theta^i$ is

$$U^i(x^i; \theta^i) = \sum_{y^i \in Y^i} Q^i_{\theta^i}(y^i \mid x^i) u^i(x^i, y^i).$$

A belief $\mu^i \in \Delta(\Theta^i)$ is player $i$'s probability measure over her parameters. Given $\mu^i$, player $i$'s best-response correspondence is

$$\mathrm{BR}^i(\mu^i) = \left\{ \tau^i \in \Sigma^i : \operatorname{supp}(\tau^i) \subseteq \arg\max_{x^i \in X^i} \int_{\Theta^i} U^i(x^i; \theta^i)\, \mu^i(d\theta^i) \right\}.$$

The belief as a whole supports player $i$'s strategy. If $\tau^i$ is mixed, one belief $\mu^i$ must make every action in its support optimal. Thus, statistical fit depends only on the conditional distributions following played actions, whereas optimality compares all actions and may depend on predictions following actions that are not played.

**Definition 1** (Berk–Nash equilibrium)**.** A strategy profile $\sigma \in \Sigma$ is a Berk–Nash equilibrium if, for every player $i \in I$, there exists a belief $\mu^i \in \Delta(\Theta^i)$ such that

$$\mu^i\big(\Theta^i(\sigma)\big) = 1 \qquad \text{and} \qquad \sigma^i \in \mathrm{BR}^i(\mu^i).$$

A belief satisfying these two conditions is a supporting belief for player $i$ at $\sigma$. Let

$$\mathcal{B}^{i,0}(\sigma) \equiv \big\{\mu^i \in \Delta(\Theta^i) : \mu^i\big(\Theta^i(\sigma)\big) = 1,\ \sigma^i \in \mathrm{BR}^i(\mu^i)\big\}$$

denote the set of player $i$'s supporting beliefs for $\sigma$, and let

$$E^0 \equiv \big\{\sigma \in \Sigma : \mathcal{B}^{i,0}(\sigma) \neq \varnothing \text{ for every } i \in I\big\}$$

denote the set of Berk–Nash equilibria.

At a Berk–Nash equilibrium, each player's strategy is optimal under a belief concentrated on the parameters that best fit her own action–consequence data. The belief $\mu^i$ is over player $i$'s subjective models. It does not describe player $j$'s belief and imposes no requirement that the strategy $\widehat{\sigma}^j$ appearing in a parameter satisfy player $j$'s fitting or optimality conditions.

We maintain the following regularity conditions.

**Assumption 1** (Regularity)**.** For every player $i \in I$:

(i) $\Pi^i$ is nonempty and compact;

(ii) for every $\theta^i \in \Theta^i$, there exists a sequence $(\theta^i_n)_n \subseteq \Theta^i$ converging to $\theta^i$ such that

$$Q^i_{\theta^i_n}(y^i \mid x^i) > 0$$

for every $x^i \in X^i$ and every $y^i \in Y^i$ satisfying $y^i = f^i(x^i, x^j, \omega)$ for some $(x^j, \omega) \in X^j \times \Omega$.

Part (ii) allows parameters that assign zero probability to some consequences, but requires them to be approximable by parameters assigning positive probability to every feasible consequence. Under Assumption 1, $\Theta^i$ is compact, $K^i(\sigma, \cdot)$ attains a finite minimum, and $\Theta^i(\sigma)$ is nonempty and compact at every profile. The finite-environment existence result of Esponda and Pouzo (2016, Theorem 1) then gives

$$E^0 \neq \varnothing.$$

**Role of the decomposition.** For Berk–Nash equilibrium, a parameter $\theta^i$ matters only through the family of conditional distributions $\big(Q^i_{\theta^i}(\cdot \mid x^i)\big)_{x^i \in X^i}$. These distributions determine both the parameter's statistical fit and the expected payoff assigned to each action. Hence Berk–Nash equilibrium does not distinguish between two parameters that induce the same family, even if their state-distribution and strategy components differ. Our decomposition gives those distributions a structural interpretation, but it adds no further restriction to Berk–Nash equilibrium.

The two components play separate roles once rationality is imposed on the behavior attributed to player $j$. Given $(\pi^i, \widehat{\sigma}^j)$ and player $i$'s strategy $\sigma^i$, player $i$ can read $(\pi^i, \sigma^i, \widehat{\sigma}^j)$ as a hypothesis about the situation faced by player $j$. In this hypothesis, Nature draws the state according to $\pi^i$, player $i$ uses $\sigma^i$, and player $j$ is attributed $\widehat{\sigma}^j$. The pair $(\pi^i, \sigma^i)$ determines the distribution of player $j$'s consequences following each action, while $\widehat{\sigma}^j$ determines which actions she takes and hence which consequences she observes.

Player $i$ can then ask whether $\widehat{\sigma}^j$ makes sense from player $j$'s point of view: could a player facing this environment, observing the consequences following her actions, and choosing optimally given the belief she forms from those observations behave according to $\widehat{\sigma}^j$? A parameter may fit player $i$'s observations while attributing to player $j$ behavior that is not rational given player $j$'s own observations. The next section formalizes this requirement as cross-player Bayesian rationality.

# 4 Cross-Player Bayesian Rationality

Berk–Nash equilibrium disciplines each player's belief through the data generated by her actions and requires her strategy to be optimal under that belief. A parameter $(\pi^i, \widehat{\sigma}^j)$ held by player $i$ also attributes a strategy to player $j$. We ask whether $\widehat{\sigma}^j$ satisfies the same fitting-and-optimality condition in the represented problem generated by that parameter and player $i$'s strategy. For a fitted parameter $(\rho, \widehat{\sigma}^j)$, the state distribution $\rho$, player $i$'s strategy, and the consequence function determine the conditional distributions of consequences in player $j$'s represented decision problem. Player $j$ evaluates these distributions using her own parameter space and a belief over her own parameters.

## 4.1 The represented decision problem

Fix distinct players $i$ and $j$, a state distribution $\rho \in \Delta(\Omega)$, and strategies $\tau^i \in \Sigma^i$ and $\tau^j \in \Sigma^j$. Suppose Nature draws the state according to $\rho$, player $i$ uses $\tau^i$, and player $j$ is attributed the strategy $\tau^j$. Conditional on action $x^j$, player $j$'s distribution of consequences is $Q^j_{\rho,\tau^i}(\cdot \mid x^j)$. The strategy $\tau^j$ determines the frequency with which these conditional distributions enter her fitting problem. Player $j$ evaluates these data using parameters $\theta^j = (\pi^j, \widehat{\tau}^i) \in \Theta^j$. The pair $(\rho, \tau^i)$ determines the data she is represented as observing, whereas $\theta^j$ determines one family of conditional distributions available in her subjective model. Neither $\pi^j = \rho$ nor $\widehat{\tau}^i = \tau^i$ is required.

For the data generated by $(\rho, \tau^i, \tau^j)$, define player $j$'s weighted Kullback–Leibler criterion and fitting set by

$$K^j_\rho(\tau^i, \tau^j; \theta^j) = \sum_{x^j \in X^j} \tau^j(x^j) \sum_{y^j \in Y^j} Q^j_{\rho,\tau^i}(y^j \mid x^j) \ln \frac{Q^j_{\rho,\tau^i}(y^j \mid x^j)}{Q^j_{\theta^j}(y^j \mid x^j)}$$

and

$$\Theta^j_\rho(\tau^i, \tau^j) = \arg\min_{\theta^j \in \Theta^j} K^j_\rho(\tau^i, \tau^j; \theta^j).$$

Thus $\Theta^j_\rho(\tau^i, \tau^j)$ contains the parameters that best fit the consequences player $j$ would observe when she uses $\tau^j$ in the represented environment. Lemma 6 in Appendix A.1 shows that, under Assumption 1, this set is nonempty and compact and the fitting correspondence has a closed graph.

**Definition 2** (Berk–Nash strategy correspondence)**.** For every $\rho \in \Delta(\Omega)$ and $\tau^i \in \Sigma^i$, define

$$\mathcal{R}^{j,1}(\rho, \tau^i) = \left\{ \tau^j \in \Sigma^j : \exists \nu^j \in \Delta(\Theta^j) \text{ such that } \nu^j \left( \Theta^j_\rho(\tau^i, \tau^j) \right) = 1 \text{ and } \tau^j \in \mathrm{BR}^j(\nu^j) \right\}.$$

The belief $\nu^j$ belongs to player $j$ in this represented problem. The belief must be concentrated on the parameters that best fit her represented data and support her strategy $\tau^j$; if $\tau^j$ mixes, it must make every action in the support optimal. Different represented problems may use different beliefs. The definition follows the logic of rationalizability, but allows only beliefs concentrated on parameters that best fit the represented data.[7]

**Lemma 1** (Existence in represented decision problems)**.** *Under Assumption 1, for every pair of distinct players $i$ and $j$, every $\rho \in \Delta(\Omega)$, and every $\tau^i \in \Sigma^i$,*

$$\mathcal{R}^{j,1}(\rho, \tau^i) \neq \varnothing.$$

The lemma guarantees at least one supported strategy in every represented decision problem. It does not imply that the strategy $\widehat{\sigma}^j$ attributed by a particular parameter $(\rho, \widehat{\sigma}^j)$ belongs to $\mathcal{R}^{j,1}(\rho, \sigma^i)$. The proof is in Appendix A.1.

**Proposition 2** (Equivalent formulation of Berk–Nash equilibrium)**.** *A strategy profile $\sigma$ is a Berk–Nash equilibrium if and only if, for every player $k \in I$, letting $\ell \neq k$ denote the other player,*

$$\sigma^k \in \mathcal{R}^{k,1}(p, \sigma^\ell).$$

*Proof.* At the actual profile, the represented problem for player $k$ is her own decision problem. For every $x^k \in X^k$,

$$Q^k_{p,\sigma^\ell}(\cdot \mid x^k) = Q^k_\sigma(\cdot \mid x^k), \qquad \Theta^k_p(\sigma^\ell, \sigma^k) = \Theta^k(\sigma).$$

Definition 2 therefore gives $\sigma^k \in \mathcal{R}^{k,1}(p, \sigma^\ell)$ exactly when $\mathcal{B}^{k,0}(\sigma) \neq \varnothing$. Applying this to both players proves the result. $\square$

The proposition characterizes the Berk–Nash equilibrium condition from Definition 1 using the same correspondence that will evaluate attributed behavior. The equivalence does not require $p \in \Pi^k$: the objective state distribution generates player $k$'s data, while minimization remains over her parameter space $\Theta^k$.

[7] For related belief-restricted rationalizability arguments, see Bernheim (1984); Pearce (1984); Rubinstein and Wolinsky (1994); Battigalli and Siniscalchi (2003). The present restriction uses Kullback–Leibler fit in the represented problem. We do not use the term *Berk–Nash rationalizability*, which Esponda and Pouzo (2025) use for a different concept.

### 4.2 The first cross-player restriction

Fix a profile $\sigma$ and a fitted parameter $(\rho, \widehat{\sigma}^j) \in \Theta^i(\sigma)$. This parameter is one hypothesis held by player $i$: it specifies the state distribution $\rho$ and attributes the strategy $\widehat{\sigma}^j$ to player $j$; it is not player $j$'s belief. Together with $\sigma^i$, it determines the represented problem defined above for player $j$.

**Definition 3** (Cross-player Bayesian rationality)**.** The fitted parameter $(\rho, \widehat{\sigma}^j)$ satisfies *cross-player Bayesian rationality* at $\sigma$ if

$$\widehat{\sigma}^j \in \mathcal{R}^{j,1}(\rho, \sigma^i).$$

Define

$$\Theta^{i,1}(\sigma) = \left\{(\rho, \widehat{\sigma}^j) \in \Theta^i(\sigma) : \widehat{\sigma}^j \in \mathcal{R}^{j,1}(\rho, \sigma^i)\right\}$$

as the set of player $i$'s fitted parameters satisfying this condition. [8]

**Definition 4** (Cross-player Berk–Nash equilibrium)**.** For each player $i$, define

$$\mathcal{B}^{i,1}(\sigma) = \left\{\mu^i \in \mathcal{B}^{i,0}(\sigma) : \mu^i\big(\Theta^{i,1}(\sigma)\big) = 1\right\}.$$

The profile is a *cross-player Berk–Nash equilibrium* if $\mathcal{B}^{i,1}(\sigma) \neq \varnothing$ for both players. Define the set of all such profiles

$$E^1 = \{\sigma \in \Sigma : \mathcal{B}^{i,1}(\sigma) \neq \varnothing \text{ for every } i \in I\}$$

**Proposition 3.** *The first cross-player restriction refines Berk–Nash equilibrium:*

$$E^1 \subseteq E^0.$$

The proof is immediate from Definition 4.

For each player $i$, one belief $\mu^i$ must support $\sigma^i$ and assign probability one to $\Theta^{i,1}(\sigma)$. For $\mu^i$-almost every parameter $(\rho, \widehat{\sigma}^j)$, a belief $\nu^j_{\rho,\widehat{\sigma}^j}$ belonging to player $j$ in the corresponding represented problem must be concentrated on $\Theta^j_\rho(\sigma^i, \widehat{\sigma}^j)$ and support $\widehat{\sigma}^j$. These represented beliefs may differ across parameters. A point belief at an individual parameter in $\operatorname{supp}\mu^i$ need not support $\sigma^i$; the belief $\mu^i$ as a whole must do so. Player $i$ first minimizes the Kullback–Leibler criterion over $\Theta^i$, obtaining $\Theta^i(\sigma)$, and then restricts which minimizers can receive probability. A parameter with greater loss does not replace a minimizer that fails cross-player Bayesian rationality. The set $\mathcal{B}^{i,1}(\sigma)$ contains the beliefs that already support $\sigma^i$ and assign probability one to $\Theta^{i,1}(\sigma)$; it is not obtained by conditioning an arbitrary belief in $\mathcal{B}^{i,0}(\sigma)$. A nonempty $\Theta^{i,1}(\sigma)$ need not contain parameters on which a belief supports $\sigma^i$. In the coordination example, this is why $(L, L)$ does not belong to $E^1$ even though $\Theta^{i,1}(L, L) \neq \varnothing$; at $(R, R)$, only the point belief on the truthful parameter remains.

[8] Under Assumption 1, $\Theta^{i,1}(\sigma)$ is compact. Appendix A.2 gives the proof.

### 4.3 Recursive cross-player Bayesian rationality

The first cross-player restriction does not restrict the parameters used by a belief that supports $\widehat{\sigma}^j$ in player $j$'s represented problem. We now require those parameters to attribute strategies that satisfy the preceding condition in the next represented problem.

**Definition 5** (Recursive cross-player Bayesian rationality)**.** Starting from $\mathcal{R}^{i,1}$ in Definition 2, for every integer $n \geq 1$, every pair of distinct players $i$ and $j$, every $\rho \in \Delta(\Omega)$, and every $\tau^j \in \Sigma^j$, define

$$\mathcal{R}^{i,n+1}(\rho, \tau^j) = \left\{ \tau^i \in \Sigma^i : \begin{array}{l} \exists \nu^i \in \Delta(\Theta^i) \text{ such that } \tau^i \in \mathrm{BR}^i(\nu^i), \text{ and} \\ \nu^i\big(\big\{(q, \widehat{\tau}^j) \in \Theta^i_\rho(\tau^j, \tau^i) : \widehat{\tau}^j \in \mathcal{R}^{j,n}(q, \tau^i)\big\}\big) = 1 \end{array} \right\}.$$

Starting from $\Theta^{i,1}(\sigma)$, $\mathcal{B}^{i,1}(\sigma)$, and $E^1$ in Definitions 3 and 4, for every integer $n \geq 2$, every $\sigma \in \Sigma$, and for every distinct players $i$ and $j$, define

$$\Theta^{i,n}(\sigma) = \big\{(\rho, \widehat{\sigma}^j) \in \Theta^i(\sigma) : \widehat{\sigma}^j \in \mathcal{R}^{j,n}(\rho, \sigma^i)\big\},$$

$$\mathcal{B}^{i,n}(\sigma) = \big\{\mu^i \in \mathcal{B}^{i,0}(\sigma) : \mu^i\big(\Theta^{i,n}(\sigma)\big) = 1\big\}.$$

For every integer $n \geq 2$, set

$$E^n = \big\{\sigma \in \Sigma : \mathcal{B}^{i,n}(\sigma) \neq \varnothing \text{ for every } i \in I\big\}.$$

Finally, for every pair of distinct players $i$ and $j$, every $\rho \in \Delta(\Omega)$, and every $\tau^j \in \Sigma^j$, define

$$\mathcal{R}^{i,\infty}(\rho, \tau^j) = \bigcap_{n \geq 1} \mathcal{R}^{i,n}(\rho, \tau^j).$$

Set

$$E^\infty = \big\{\sigma \in \Sigma : \sigma^i \in \mathcal{R}^{i,\infty}(p, \sigma^j) \text{ for every } i \in I\big\}.$$

Because the represented fitting problem at $(p, \sigma^j, \sigma^i)$ coincides with player $i$'s fitting problem at the actual profile, for every integer $n \geq 0$,

$$\sigma \in E^n \quad \Longleftrightarrow \quad \sigma^i \in \mathcal{R}^{i,n+1}(p, \sigma^j) \text{ for every } i \in I.$$

For $n = 0$, this is Proposition 2; for $n \geq 1$, it follows from the definitions of $\Theta^{i,n}(\sigma)$ and $\mathcal{B}^{i,n}(\sigma)$. The correspondence $\mathcal{R}^{i,1}$ applies fitting and optimality to player $i$ in the current problem, while $E^n$ adds $n$ cross-player applications. Both actual players are assessed at the same value of $n$.

In $\mathcal{R}^{i,n+1}(\rho, \tau^j)$, the belief $\nu^i$ belongs to player $i$ in the represented problem determined by $(\rho, \tau^j, \tau^i)$. It supports $\tau^i$ and is concentrated on player $i$'s best-fitting parameters $(q, \widehat{\tau}^j)$ for which $\widehat{\tau}^j \in \mathcal{R}^{j,n}(q, \tau^i)$. The beliefs supporting that condition belong to player $j$ in the corresponding represented problems and may differ across parameters. At every application, $\Theta^i_\rho(\tau^j, \tau^i)$ remains the fitting set obtained from the full parameter space $\Theta^i$; the recursion only restricts which minimizers can receive probability.

**Lemma 4** (Further applications refine earlier ones)**.** *For every integer $n \geq 1$, every pair of distinct players $i$ and $j$, every $\rho \in \Delta(\Omega)$, and every $\tau^j \in \Sigma^j$,*

$$\mathcal{R}^{i,n}(\rho, \tau^j) \supseteq \mathcal{R}^{i,n+1}(\rho, \tau^j).$$

*For the same $n$, every $\sigma \in \Sigma$, and every player $i$,*

$$\Theta^{i,n}(\sigma) \supseteq \Theta^{i,n+1}(\sigma), \qquad \mathcal{B}^{i,n}(\sigma) \supseteq \mathcal{B}^{i,n+1}(\sigma),$$

*and every profile in $E^{n+1}$ also belongs to $E^n$.*

*Proof.* A belief satisfying the conditions for $\tau^i \in \mathcal{R}^{i,2}(\rho, \tau^j)$ is concentrated on a subset of $\Theta^i_\rho(\tau^j, \tau^i)$ and therefore also satisfies the conditions for $\tau^i \in \mathcal{R}^{i,1}(\rho, \tau^j)$. Now suppose that $\mathcal{R}^{j,n}(q, \tau^i)$ includes $\mathcal{R}^{j,n+1}(q, \tau^i)$ for every $q$ and $\tau^i$. Any belief satisfying the conditions for $\mathcal{R}^{i,n+2}(\rho, \tau^j)$ then also satisfies the conditions for $\mathcal{R}^{i,n+1}(\rho, \tau^j)$. Repeating this argument proves the first statement. The definitions of $\Theta^{i,n}(\sigma)$, $\mathcal{B}^{i,n}(\sigma)$, and $E^n$ give the remaining statements. □

Each additional application may remove strategies and profiles but cannot add a profile excluded at an earlier step. Together with Proposition 3, the lemma gives

$$E^0 \supseteq E^1 \supseteq E^2 \supseteq \cdots .$$

The actual-profile relation and the definition of $\mathcal{R}^{i,\infty}$ also give

$$E^\infty = \bigcap_{n \geq 0} E^n.$$

Thus every profile in $E^\infty$ belongs to $E^n$ for every finite $n$. These are behavioral restrictions on fitting and choice in represented decision problems; they do not specify epistemic types, a common prior over types, or literal knowledge of rationality.

# 5 Specification and Identification

Can the first cross-player restriction distinguish hypotheses that generate the same observations for the player who holds them, and thereby identify player $i$'s belief supporting her strategy at a fixed profile? Truth admission, data reproduction, and identification answer different questions. For distinct players $i$ and $j$, player $i$ admits the truth when $p \in \Pi^i$. Her subjective model reproduces her data at a profile $\sigma$ when some $\theta^i \in \Theta^i$ satisfies

$$Q^i_{\theta^i}(\cdot \mid x^i) = Q^i_\sigma(\cdot \mid x^i) \qquad \text{for every } x^i \in \operatorname{supp}(\sigma^i).$$

Truth admission guarantees reproduction at every profile because $(p, \sigma^j)$ reproduces player $i$'s conditional distributions of consequences after every action. The converse need not hold: a false state distribution combined with an incorrect attributed strategy may reproduce the same data. Under Assumption 1, reproduction at $\sigma$ is equivalent to zero minimum Kullback–Leibler loss, in which case every parameter in $\Theta^i(\sigma)$ reproduces the conditional distributions following player $i$'s played actions. Neither truth admission nor reproduction by itself guarantees identification of either parameter component or of player $i$'s supporting belief.

**Definition 6** (Observational equivalence)**.** Fix a profile $\sigma$ and player $i$. Two parameters $\theta^i, \widetilde{\theta}^i \in \Theta^i$ are *observationally equivalent at* $\sigma$ for player $i$, written $\theta^i \sim^i_\sigma \widetilde{\theta}^i$, if

$$Q^i_{\theta^i}(\cdot \mid x^i) = Q^i_{\widetilde{\theta}^i}(\cdot \mid x^i) \qquad \text{for every } x^i \in \operatorname{supp}(\sigma^i).$$

Observationally equivalent parameters have the same Kullback–Leibler loss at $\sigma$. Hence, if $\theta^i \in \Theta^i(\sigma)$ and $\widetilde{\theta}^i \sim^i_\sigma \theta^i$, then $\widetilde{\theta}^i \in \Theta^i(\sigma)$, including when the minimum loss is positive. Player $i$'s actual-profile fitting problem cannot distinguish the two hypotheses. They are hypotheses held by player $i$, not beliefs of player $j$.

The first cross-player restriction may nevertheless distinguish them. Fix a fitted hypothesis $(\rho, \widehat{\sigma}^j) \in \Theta^i(\sigma)$. It belongs to $\Theta^{i,1}(\sigma)$ exactly when $\widehat{\sigma}^j \in \mathcal{R}^{j,1}(\rho, \sigma^i)$. The state distribution $\rho$, player $i$'s strategy, and player $j$'s consequence function generate the conditional distributions of consequences in player $j$'s represented problem, while $\widehat{\sigma}^j$ determines which actions she samples. A belief belonging to player $j$ in that problem must be concentrated on her best-fitting parameters and support $\widehat{\sigma}^j$. Observationally equivalent hypotheses for player $i$ may therefore generate different conditional distributions for player $j$, attribute different strategies to her, or both. A different attributed strategy changes both the sampling weights in player $j$'s fitting criterion and the strategy that her belief must support. Hence one hypothesis may belong to $\Theta^{i,1}(\sigma)$ while the other does not. This restriction adds no observation and does not minimize player $i$'s criterion again; it restricts which of her original best-fitting parameters may receive probability in a belief supporting $\sigma^i$.

**Definition 7** (Complete supporting-belief identification)**.** Fix a profile $\sigma$ and player $i$ such that $\mathcal{B}^{i,1}(\sigma) \neq \varnothing$. Player $i$'s supporting belief is *completely identified after the first cross-player application* if $\mathcal{B}^{i,1}(\sigma)$ is a singleton.

Complete supporting-belief identification requires uniqueness of the entire probability measure over player $i$'s parameters. On-path supporting-belief identification is weaker: it requires only that all supporting beliefs induce the same distribution over likelihood-relevant predictions following played actions. No result about that weaker property is stated here. Conditional on $\mathcal{B}^{i,1}(\sigma) \neq \varnothing$, a singleton $\Theta^{i,1}(\sigma)$ implies complete supporting-belief identification, but the converse can fail because a unique supporting belief may assign positive probability to several parameters. Belief identification is conditional on the fixed profile and does not imply that the strategy profile is unique.

The coordination game in Example 3 illustrates both profile rejection and supporting-belief identification without a second calculation here. At $(L, L)$, each player's set $\Theta^{i,1}(L, L)$ is nonempty, but no belief concentrated on that set supports $L$; this is behavioral rejection, not identification. At $(R, R)$, $\mathcal{B}^{i,1}(R, R) = \{\delta_{(1/2,0)}\}$ for each player $i$. The first cross-player restriction leaves behavior unchanged at this profile and completely identifies each actual player's supporting belief. The uniqueness statement concerns the actual players' supporting beliefs and makes no uniqueness claim about beliefs in other represented problems.

Supporting-belief identification is a fixed-profile question. Existence in $E^\infty$ instead asks whether at least one common profile satisfies both players' restrictions at every finite order. The binary analysis turns to that question under arbitrary feedback.

# 6 Existence in Binary Games

Can the players' subjective models support one common profile when fitting and optimality are imposed through every represented problem? With two actions, each support question is determined by one expected payoff difference. The recursive restriction determines which best-fitting hypotheses may receive probability, while optimality determines the sign their average payoff difference must have. We impose no restriction on feedback, so the state-distribution and attributed-strategy components of a hypothesis may jointly explain the observed consequences.

## 6.1 Payoff-gap geometry

Suppose $\Omega = X^1 = X^2 = \{0, 1\}$, write $p_0 = p(1) \in (0, 1)$, and identify a state distribution with its probability of state 1. Maintain Assumption 1. For a parameter $(q, \widehat{\tau}^j) \in \Theta^i$, define player $i$'s payoff difference between actions 1 and 0 by

$$\Delta^i(q, \widehat{\tau}^j) = U^i(1; (q, \widehat{\tau}^j)) - U^i(0; (q, \widehat{\tau}^j)).$$

For a strategy $\tau^i \in \Sigma^i$, define its best-response region in payoff-gap space by

$$\mathcal{I}^i(\tau^i) = \begin{cases} (-\infty, 0], & \tau^i(1) = 0, \\ \{0\}, & 0 < \tau^i(1) < 1, \\ [0, \infty), & \tau^i(1) = 1. \end{cases}$$

An expected payoff difference lies in $\mathcal{I}^i(\tau^i)$ exactly when every action in the support of $\tau^i$ is optimal.

Fix distinct players $i$ and $j$, a represented state probability $\rho$, and a strategy pair $(\tau^i, \tau^j)$. The payoff differences attainable under beliefs satisfying the fitting restriction form

$$\mathcal{G}^{i,0}_\rho(\tau^j, \tau^i) = \operatorname{co}\left\{\Delta^i(q, \widehat{\tau}^j) : (q, \widehat{\tau}^j) \in \Theta^i_\rho(\tau^j, \tau^i)\right\}.$$

For every integer $n \geq 1$, the payoff differences attainable after $n$ cross-player applications form

$$\mathcal{G}_\rho^{i,n}(\tau^j, \tau^i) = \operatorname{co}\left\{ \Delta^i(q, \widehat{\tau}^j) : \begin{array}{l} (q, \widehat{\tau}^j) \in \Theta_\rho^i(\tau^j, \tau^i), \\ \widehat{\tau}^j \in \mathcal{R}^{j,n}(q, \tau^i) \end{array} \right\},$$

where $\operatorname{co} \varnothing = \varnothing$. The convexification is over payoff differences generated by complete parameters, not over their state-distribution components separately.

Under Assumption 1, the parameter set entering each finite-order convex hull is compact. The value of $\int \Delta^i(q, \widehat{\tau}^j)\, d\nu^i(q, \widehat{\tau}^j)$ under a belief concentrated on that set can therefore be any point in $\mathcal{G}_\rho^{i,n}(\tau^j, \tau^i)$. The recursive definition consequently gives

$$\tau^i \in \mathcal{R}^{i,n+1}(\rho, \tau^j) \iff \mathcal{G}_\rho^{i,n}(\tau^j, \tau^i) \cap \mathcal{I}^i(\tau^i) \neq \varnothing \qquad (n \geq 0). \tag{1}$$

The belief $\nu^i$ is player $i$'s belief in this represented problem. One belief must both be concentrated on the indicated best-fitting parameters and support the strategy $\tau^i$. Equation (1) assumes neither feedback identification nor a zero minimum Kullback–Leibler loss.

At the actual state distribution, the relation between $E^n$ and $\mathcal{R}^{i,n+1}$ yields

$$\sigma \in E^n \iff \mathcal{G}_{p_0}^{i,n}(\sigma^j, \sigma^i) \cap \mathcal{I}^i(\sigma^i) \neq \varnothing \quad \text{for both players } i \qquad (n \geq 0). \tag{2}$$

A profile $\sigma$ is in $E^\infty$ exactly when the two intersections in (2) are nonempty at every finite order.

Lemma 1 guarantees that $\mathcal{R}^{i,1}(\rho, \tau^j)$ is nonempty in every represented environment. It does not imply that this correspondence contains a prescribed strategy, that a higher-order correspondence is nonempty, or that the two actual players' conditions hold at one common profile. As $n$ increases, the parameter set defining $\mathcal{G}_\rho^{i,n}$ can only shrink, while $\mathcal{I}^i(\tau^i)$ remains fixed.

There are two distinct routes for a proposed strategy to fail. First, no best-fitting parameter may attribute a strategy satisfying the preceding-order restriction, in which case $\mathcal{G}_\rho^{i,n}$ is empty. Second, such parameters may exist, but every payoff-gap average they generate lies outside the proposed strategy's best-response region. Thus, for every integer $n \geq 1$,

$$E^n = \varnothing \iff \text{every } \sigma \in E^{n-1} \text{ has some player } i \text{ for whom } \mathcal{G}_{p_0}^{i,n}(\sigma^j, \sigma^i) \cap \mathcal{I}^i(\sigma^i) = \varnothing.$$

If this occurs at one finite order, nestedness implies that every subsequent $E^m$, $m > n$, and $E^\infty$ are empty.

### 6.2 Conditions for existence

For every $(r_1, r_2) \in [0,1]^2$, let $\Gamma(r_1, r_2)$ denote the strategic game in which player $i$ evaluates her actions using state probability $r_i$. Its Nash equilibrium set is

$$N(\Gamma(r_1, r_2)) = \left\{ \sigma \in \Sigma : \operatorname{supp}(\sigma^i) \subseteq \arg\max_{x^i \in X^i} U^i(x^i; (r_i, \sigma^j)) \text{ for each player } i \right\}.$$

When $r_1 \neq r_2$, this is the no-private-information specialization of Nash equilibrium with diverse priors in Dekel, Fudenberg, and Levine (2004); at $(r_1, r_2) = (p_0, p_0)$, the prior is common and correct.

**Proposition 5** (Truth admission and mutually fitted hypotheses)**.** *Under Assumption 1, the following statements hold.*

(i) *If $p_0 \in \Pi^i$ for both players, then*

$$N(\Gamma(p_0, p_0)) \subseteq E^\infty,$$

*and therefore $E^\infty \neq \varnothing$.*

(ii) *More generally, fix $q_i \in \Pi^i$ for each player and $\sigma \in N(\Gamma(q_1, q_2))$. If, for each player $i$ and the other player $j$,*

$$(q_i, \sigma^j) \in \Theta^i_{p_0}(\sigma^j, \sigma^i) \cap \Theta^i_{q_j}(\sigma^j, \sigma^i),$$

*then $\sigma \in E^\infty$.*

*Proof.* For part (ii), fix player $i$. The point belief on $(q_i, \sigma^j)$ is player $i$'s belief and supports $\sigma^i$ by Nash optimality in $\Gamma(q_1, q_2)$. The first fitting condition places $(q_i, \sigma^j)$ among the best-fitting parameters in player $i$'s actual problem. The second places the same parameter among the best-fitting parameters in her represented problem with state probability $q_j$. The point belief is therefore concentrated on best-fitting parameters in both problems and establishes the first-order condition in each. Suppose, simultaneously for both players, that the corresponding strategies satisfy the order-$n$ condition in these two problems. In each order-$(n+1)$ problem, the parameter $(q_i, \sigma^j)$ attributes $\sigma^j$, which satisfies player $j$'s order-$n$ condition in the next represented problem. The same point belief then establishes the order-$(n+1)$ condition. Induction gives the conclusion at every finite order. For part (i), set $q_1 = q_2 = p_0$. At an objective Nash equilibrium, $(p_0, \sigma^j)$ reproduces player $i$'s conditional distributions of consequences after every action and therefore has zero Kullback–Leibler loss in both relevant problems. Part (ii) applies. The finite game $\Gamma(p_0, p_0)$ has a mixed Nash equilibrium, which proves nonemptiness. □

In Proposition 5(ii), the first fitting condition concerns player $i$'s actual data. The second concerns her data in the represented problem generated by player $j$'s selected state distribution. In both problems, $(q_i, \sigma^j)$ is one hypothesis held by player $i$; it is not player $j$'s belief. These conditions are sufficient, not necessary. A belief over several best-fitting parameters may generate a payoff difference in $\mathcal{I}^i(\sigma^i)$ even when no pair of point beliefs satisfies part (ii).

Truth admission therefore guarantees objective Nash behavior at every finite order, but it does not imply $E^\infty = E^0$, identify either supporting belief, or exclude additional profiles. Conversely, when $E^\infty$ is empty, the conclusion is not that every represented decision problem lacks an optimal strategy. It means that no common profile can be accompanied by player-owned beliefs that satisfy fitting and optimality in all the required represented problems. Misspecification can produce this incompatibility because the parameters minimizing one player's criterion may attribute behavior that the other player's model cannot support, or because the parameters satisfying that restriction generate payoff differences inconsistent with the first player's proposed strategy.

With more than two actions, the payoff-gap interval becomes a convex set of payoff vectors and the sign region becomes the best-response region for a mixed strategy. The following section develops that general geometry together with the limiting recursive argument.

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# A Proofs and Calculations

This appendix contains the proofs and calculations used in Sections 4–6. The order follows the first appearance of each result in the main text.

## A.1 Represented fitting is well defined

**Lemma 6** (Represented fitting)**.** *Under Assumption 1, for every pair of distinct players $i$ and $j$ and every $(\rho, \tau^i, \tau^j) \in \Delta(\Omega) \times \Sigma^i \times \Sigma^j$, the function $K^j_\rho(\tau^i, \tau^j; \cdot)$ takes at least one finite value and attains a finite minimum on $\Theta^j$. Consequently, $\Theta^j_\rho(\tau^i, \tau^j)$ is nonempty and compact. The correspondence $(\rho, \tau^i, \tau^j) \mapsto \Theta^j_\rho(\tau^i, \tau^j)$ has a closed, hence compact, graph.*

*Proof.* Write $\alpha = (\rho, \tau^i, \tau^j)$ and $K^j(\alpha; \theta^j) = K^j_\rho(\tau^i, \tau^j; \theta^j)$. This is relative entropy between two finite joint distributions after terms for actions assigned zero probability by $\tau^j$ are omitted, and it is jointly lower semicontinuous. Regularity supplies a parameter assigning positive probability to every primitively feasible sampled consequence and therefore a finite-loss comparator. Compactness of $\Theta^j$ gives a finite attained minimum and a nonempty compact argmin.

For closedness, take $\alpha_n \to \alpha$, $\theta^j_n \to \theta^j$, and $\theta^j_n \in \Theta^j_{\rho_n}(\tau^i_n, \tau^j_n)$. Fix a finite-loss comparator $\vartheta^j$ at $\alpha$. The density part of Regularity supplies $\vartheta^j_m \to \vartheta^j$ with positive probability on every primitively feasible consequence. For fixed $m$, $K^j(\,\cdot\,; \vartheta^j_m)$ is continuous at $\alpha$, and $K^j(\alpha; \vartheta^j_m) \to K^j(\alpha; \vartheta^j)$. A diagonal sequence $\vartheta^j_{m(n)}$ therefore satisfies $K^j(\alpha_n; \vartheta^j_{m(n)}) \to K^j(\alpha; \vartheta^j)$. Optimality and lower semicontinuity imply

$$K^j(\alpha; \theta^j) \le \liminf_n K^j(\alpha_n; \theta^j_n) \le \limsup_n K^j(\alpha_n; \vartheta^j_{m(n)}) = K^j(\alpha; \vartheta^j).$$

This holds for every finite-loss comparator, while infinite-loss comparators cannot lower the finite minimum. Hence $\theta^j$ is a global minimizer at $\alpha$. The graph is closed and, because its ambient spaces are compact, compact. $\square$

*Proof of Lemma 1.* Fix distinct players $i$ and $j$, $\rho \in \Delta(\Omega)$, and $\tau^i \in \Sigma^i$. Consider the finite one-player decision problem with state space $\widetilde{\Omega} = \operatorname{supp}(\rho) \times \operatorname{supp}(\tau^i)$ and objective distribution $\widetilde{p}(\omega, x^i) = \rho(\omega)\tau^i(x^i)$. This distribution has full support on $\widetilde{\Omega}$. Give player $j$ the consequence function $\widetilde{f}^j(x^j; (\omega, x^i)) = f^j(x^j, x^i, \omega)$, payoff function $u^j$, and subjective model $(\Theta^j, (Q^j_{\theta^j})_{\theta^j \in \Theta^j})$.

For every $\tau^j \in \Sigma^j$, the objective conditional distribution of consequences in this problem is $Q^j_{\rho,\tau^i}$, its weighted Kullback–Leibler criterion is $K^j_\rho(\tau^i, \tau^j; \cdot)$, and its fitting set is $\Theta^j_\rho(\tau^i, \tau^j)$. Lemma 6 makes these fitting sets well defined. Assumption 1 supplies compactness and continuity of the subjective model and the required positivity for every consequence feasible in the induced problem. Hence Esponda and Pouzo (2016, Theorem 1) gives $\bar{\tau}^j \in \Sigma^j$ and $\bar{\nu}^j \in \Delta(\Theta^j)$ such that $\bar{\nu}^j(\Theta^j_\rho(\tau^i, \bar{\tau}^j)) = 1$ and $\bar{\tau}^j \in \mathrm{BR}^j(\bar{\nu}^j)$. Therefore $\bar{\tau}^j \in \mathcal{R}^{j,1}(\rho, \tau^i)$. $\square$

## A.2 Compactness of the first cross-player restriction

**Proposition 7** (Compactness at the first cross-player application)**.** *Under Assumption 1, the graph of $(\rho, \tau^i) \mapsto \mathcal{R}^{j,1}(\rho, \tau^i)$ is closed. Consequently, $\Theta^{i,1}(\sigma)$ is compact, possibly empty, for every player $i$ and profile $\sigma$.*

*Proof.* Let $(\rho_n, \tau^i_n, \tau^j_n) \to (\rho, \tau^i, \tau^j)$ with $\tau^j_n \in \mathcal{R}^{j,1}(\rho_n, \tau^i_n)$. Choose a belief $\nu^j_n$ concentrated on $\Theta^j_{\rho_n}(\tau^i_n, \tau^j_n)$ that supports $\tau^j_n$. Compactness of $\Delta(\Theta^j)$ yields a subsequence converging weakly to some $\nu^j$. The product measures $\delta_{(\rho_n, \tau^i_n, \tau^j_n)} \otimes \nu^j_n$ converge weakly to $\delta_{(\rho, \tau^i, \tau^j)} \otimes \nu^j$. The fitting graph is closed by Lemma 6; the Portmanteau theorem therefore implies that $\nu^j$ assigns probability one to $\Theta^j_\rho(\tau^i, \tau^j)$.

The strategy best-response condition is the finite collection of closed inequalities

$$\tau^j(x) \left[ \int U^j(x; \theta^j)\, d\nu^j - \int U^j(x'; \theta^j)\, d\nu^j \right] \geq 0, \qquad x, x' \in X^j.$$

Continuity of expected utility and weak convergence preserve these inequalities, so $\tau^j \in \mathrm{BR}^j(\nu^j)$. Hence $\tau^j \in \mathcal{R}^{j,1}(\rho, \tau^i)$ and the graph is closed.

For fixed $\sigma$, the fitting set $\Theta^i(\sigma)$ is compact. The set $\Theta^{i,1}(\sigma)$ is its intersection with the inverse image of the closed graph of $\mathcal{R}^{j,1}$ under $(\rho, \widehat{\sigma}^j) \mapsto (\rho, \sigma^i, \widehat{\sigma}^j)$. It is therefore compact. $\square$

## A.3 The coordination calculation

For the higher-order calculation at $(L, L)$, write

$$A = (1/2, 1), \qquad B = (2/3, 3/4), \qquad C = (1, 1/2).$$

For any represented problem, let $q$ denote its state probability and let $s$ denote the probability of $L$ attributed to the opponent. Expected payoffs are

$$U(L; (q, s)) = 3qs, \qquad U(R; (q, s)) = 2 - \frac{3}{2} q(1 - s).$$

Section 2 calculation gives $\Theta^i(L, L) = \{A, B, C\}$ and $\Theta^{i,1}(L, L) = \{A, B\}$. The remaining task is to determine when $A$ and $B$ cease to satisfy the higher-order restrictions. Thus the second application begins from the two hypotheses $A$ and $B$.

The parameter restrictions at depths two and three remain to be calculated. Consider first $A$. Its represented problem has state probability $1/2$ and attributes pure $L$ to the opponent. A belief supports $L$ in that problem only if it assigns positive probability to $C$. Under $C$, however, the next represented problem has state probability 1 and attributes probability $1/2$ to each action. Because both actions are sampled, the unique best-fitting parameter is $(1, 1)$, under which $L$ is strictly preferred to $R$. No belief concentrated on this singleton can support the attributed mixture. Hence $A \notin \Theta^{i,2}(L, L)$.

For $B$, the represented problem has state probability $2/3$ and attributes probabilities $3/4$ and $1/4$ to $L$ and $R$. Sampling both actions uniquely identifies $(2/3, 1)$. At this parameter both actions yield 2, so the attributed mixture is optimal and $B \in \Theta^{i,2}(L, L)$. The parameter $(2/3, 1)$ in turn defines a represented problem with state probability $2/3$ and attributed pure-$L$ play. Its best-fitting set is $\{A, B, C\}$. A belief can support $L$ after the first cross-player application only by assigning positive probability to $C$, because both $A$ and $B$ give a negative payoff difference between $L$ and $R$. The preceding paragraph shows that $C$ cannot satisfy the next restriction. Therefore $B \notin \Theta^{i,3}(L, L)$, and

$$\Theta^{i,2}(L, L) = \{B\}, \qquad \Theta^{i,3}(L, L) = \varnothing.$$

This proves that $A$ is excluded at depth two, whereas $B$ satisfies the depth-two restriction but is excluded at depth three.